

Establishing an epidemiological field laboratory in rural areas - potentials for public health research and interventions

The Butajira Rural Health Programme 1987-1999

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1. BACKGROUND

Basic health care and essential health research need an epidemiological perspective. This requires local data on the health situation of the whole community. Such an approach must not be solely a method of counting heads, but must also relate to cultural patterns and values. When fed back to the people who are the source of the information, epidemiological data may "talk", and increase our understanding of causes and needs. A valid health information system is also a prerequisite for the analysis of secular trends and for their delineation from societal interventions.

For low-income countries there is also a demand for *timely vital statistics*. Epidemiological surveillance represents on-going systems for the collection, analysis and interpretation of health data, which are essential to the planning, implementation and evaluation of health programmes. An effective surveillance system also implies timely dissemination of its data to those who need to know. The final link in the surveillance chain is the application of these data to prevention and disease control. Epidemiological surveillance activities are usually included as part of the health information systems of governmental institutions that have responsibility for health care.

Basic epidemiological know-how can be gathered through formal courses or textbooks, but practical experiences can only be learnt through research out in the field. Within the so-called *sandwich model* of research training, most of the trainees' time is spent on the job, researching questions that relate to local

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situations. Technology also has a part to play here; telephone, fax and e-mail, even to some of the world's most remote communities, have developed appreciably over the last few years. Fieldwork is no longer constrained by computer facilities. Field- and process-oriented teaching of epidemiology and public health also increases awareness of the community perspective among medical personnel who previously were only clinically oriented.

In developing countries, adequate and reliable health information is often lacking. There is usually no systematically organised registration of vital events. Also, national figures are of limited use for health planning at regional levels. Population studies are therefore needed.

The overall objectives of the Butajira "field laboratory" were to develop and evaluate a system for continuous registration of births and deaths, to generate valid data on fertility and mortality and to provide a population and study base for essential health research and intervention in the area. The ultimate aim of the project was therefore to provide a current epidemiological surveillance system at district level.

Specifically, this aimed to enable the assessment of fertility and mortality trends, the analysis of public health and health behavioural determinants of mortality and morbidity as well as the assessment of coverage and utilisation of health services in the area in relation to health needs of the people.

When establishing the study base in the Butajira district it was essential to get the support of the government officials and mass organisations concerned. Letters were written by higher authorities to all those concerned requesting their support and co-operation. Close links were established with government officials and mass organisations, local leaders and members of the community. Often, open-air meetings were held with members of peasants' associations, as well as with mosque and church leaders in the villages, to explain further the purpose of the project. A considerable amount of co-operation was solicited from the participants, especially their willingness to accept and respond to interviewers who would be going into their houses and asking a number of questions for months and years to come. Informed consent and participation was sought from the onset through community leaders [1].

This report is an attempt to share experiences from the first thirteen years of the Butajira Rural Health Programme (BRHP), to present some basic data on demography, to summarise results from specific applications of its epidemiological laboratory and to suggest and outline research and intervention activities departing from the major findings so far.



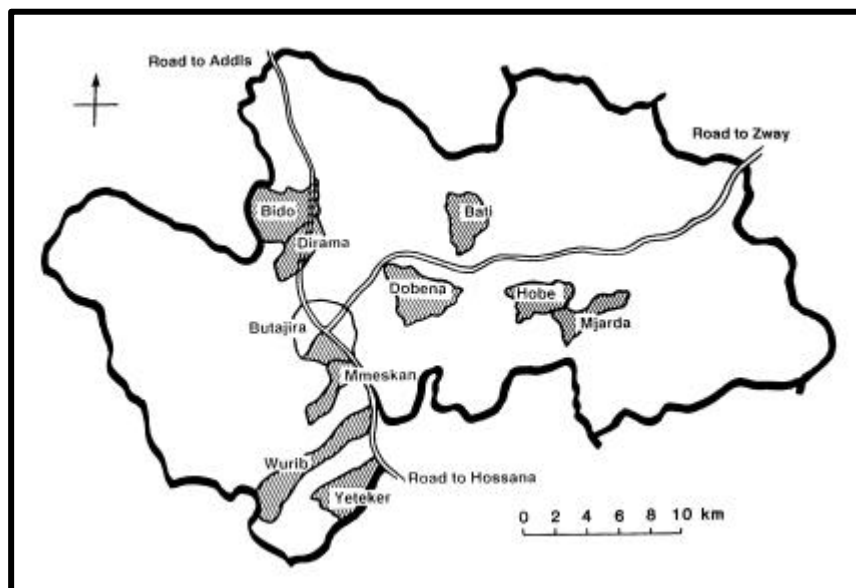
A view from the rural village

2. SETTING THE SCENE

2.1 Geography

The Butajira Rural Health Programme (BRHP) is located in Meskan and Mareko District, Gurage Zone, in the Southern Nations, Nationalities and Peoples Regional State (SNNPRG) in Ethiopia. Historically the District was part of what was known as "Gurageland". The District acquired its present name in 1954.

At the same time Butajira became the capital town of the District. The District borders the Silti District to the south, the Adamitulu District to the east, the Sabat-Bet Gurage to the west and the Sodo District to the north. The estimated size of the District is 797 km², of which Butajira town covers approximately 9 km².



Location of the study sites in the district



Sorghum field

The District lies at an average of 2,100 m above sea level, ranging from 1,750 m in the lowlands to 3,400 m in mountainous areas, which are sparsely inhabited.

Annual rainfall ranges between 900 and 1,400 mm. The main rainy season is from June to September, with "small rains" common around March and April.

The area has rich soil and farm-ing is the main mode of living. *Teff*

(a cereal staple), maize, millet, barley and legumes are the main crops. Pepper, coffee and *khat* (a mild stimulant drug) are also grown as cash crops. *Enset* (false banana) cultivation is very common and gives the main staple food in the area. Extensive crop cultivation and relatively high population density leave very little room for animal husbandry, other than draught animals.



Teff threshing

2.2 People

The Gurage people migrated from the northern part of Ethiopia in the ninth century with the expansion of the Christian kingdom southwards. Since then they have been living together with the Kushitic and Omotic people who were previously in the area. The Meskan, Mareko, Sodo, Siliti and Dobi tribal groups dominate. The Mareko are predominantly Kushitic. They were also the first to come into the area and they occupy a greater part of the land compared to the other tribes. The Siliti group, though Semitic in origin, migrated to the area from the eastern part of Ethiopia (Harar) in the sixteenth century. Several clans are identified within each tribal group.

The population of the district was given as 182,000 during the 1984 census and on the basis of the 1994 census is estimated to have grown to 257,000 by 1999. The Butajira town population has grown from about 13,000 in 1984 to 25,000 by 1999. The population structure shows a typical developing country pattern. The proportion of children under 14 years is 46% and that of people 65 years and older is 1.3% [2, 3].

The vast majority of the District population follow the Christian and Islamic religions. Currently Islam dominates, being followed by two-thirds of the District population.



Well in a lowland village



Enset (false banana plant) processing

Islam started to expand extensively in the area after the *Ahmed Gragn* invasion in the sixteenth century and subsequently through the expansion of Islamic education by very influential and feared religious leaders [4]. Sodo and Dobi tribes are predominantly Orthodox Christians while others are predominantly Moslem.

The predominant language, *Guragigna*, belongs to the Ethio-semitic language group, resembling the northern Ethiopian languages but influenced by the Kshitic language of the surroundings. The people have developed a variety of dialects of *Guragigna* within their clans. The Mareko group, however, have a distinct language, *Marekogna*, that more closely resembles the Kshitic language spoken by their neighbours, the Hadiyas.

2.3 Traditional systems

Until recently, the tribes have depended on their own traditional systems of administration to keep order and settle disputes. Clans are the basis for defining power. Their traditional rules are influenced by religion and culture, and thus differ from tribe to tribe. The establishment of a central authority in Ethiopia at the beginning of the nineteenth century started to gradually erode the power of the clan administration. The central authority commonly appointed men from other areas as governors, though sometimes were chosen. Currently, traditional rules have very limited influence and people are largely governed by the law of the country. As the national government structure developed during the twentieth century, local people have become more and more involved in the



Health post in one of the study sites

government system rather than observing traditional practices.

Historically, the area was known for its frequent civil wars. This was mainly due to the Oromo expansionary attempts from outside, and the need among tribal groups within the area to gain more land for farming and cattle grazing.

Furthermore, the custom of cutting off the genitals of enemies to testify to bravery among the Mareko, along with religious expansionism, contributed to tribal clashes in the area. These tragic conflicts have subsided since the 1940s when the Marekos were defeated by an alliance of Gurage tribes [4].



Market in Butajira: chicken trading

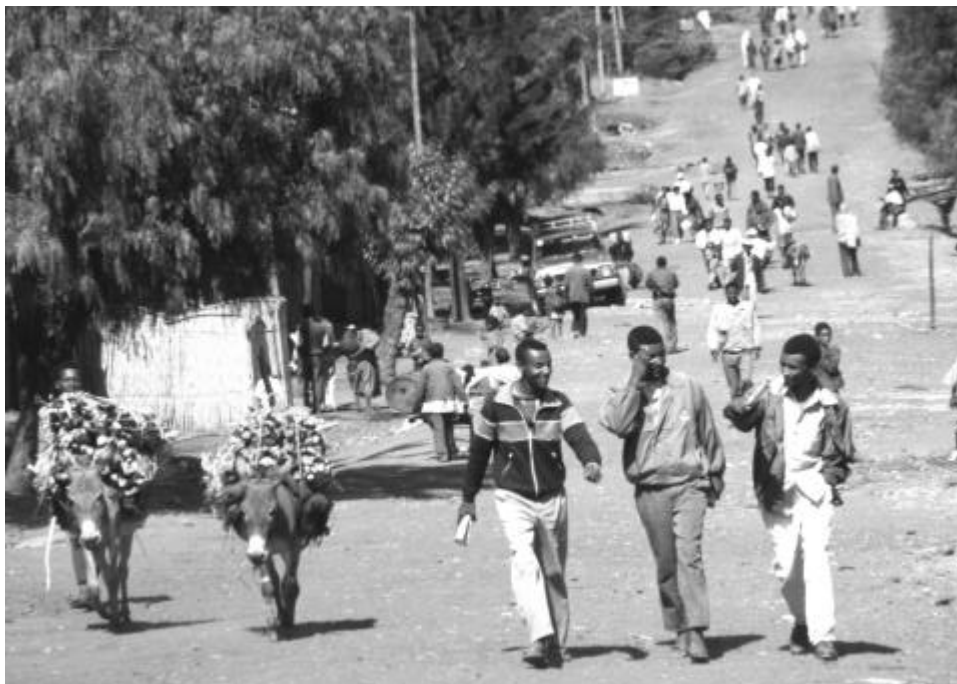
2.4 Butajira town

The town is located at 2,100 m above sea level. The municipality administers 9 km² of land to belong to the town. The town has a temperate (*woyna-dega*) climate which favours human habitation and a wide range of cultivation. It is located 130 km south of Addis Ababa and 50 km to the west of Zway town in the Rift Valley.

Butajira town has grown from a small settlement that was established by the Shoan military forces - the *Naftegna* - over hundred years ago. The soldiers were known as *Kita-Bal* as they were paid only in kind. The official establishment of Butajira town is dated as 1929. The town area was earlier occupied by the Mareko tribe but subsequently annexed by the Meskan tribe after they conquered the Marekos in war [4].

The existence of the River Eresha in the north of the town along the road to Zway and the River Rinza to the south made the location suitable for settlement because of the supply of water. This strategic location of the town and its service as market area for surrounding communities contributed to its growth.

The growth of the town was marked by the appearance of a cluster of thatched roof huts. The market then was very small and was used only by the local people. The main form of trade was barter and using *amole* - a type of rock salt. Improvements in road communications and the production of crops like *teff* and pepper encouraged the expansion of the market and thus of the town.



Main road in Butajira town

The growth of the market attracted more people from the surroundings and merchants from other places for settlement. The main items of trade in earlier times were slaves, cattle, butter, hides and skins and *bulluko* (heavy cotton blankets) that came from as far as the southern border of the country. In 1923 the Government of Ethiopia decreed a law prohibiting slave trade. However, the slave trade was not completely abolished until the Italian occupation of the town in 1936.

The naming of the town has an interesting story. Butajira means in Oromigna "is Buta there?". This originated from the presence of a *shifta* (bandit) by the name of Buta. The man was originally from the Sebat-Bet Gurage tribe, but expelled because of his bad behaviour. After being rejected by tribes around the town as well, he settled himself by

the hill in the eastern part of the town, where St. Mary's church is now located. Buta then continued to loot and even kill the Oromo traders and travellers coming to the town. Thus, people coming to the town from that direction, being very scared of him, before approaching the place started to ask inhabitants whether he had been seen around by saying "*Buta-Jira?*" [4]

The town's population continued to increase in size after its establishment in 1929. The composition of the population also changed subsequently due to an influx of traders looking for better markets. The local settlement of the *Amhara Neftegna* (soldiers) provided peace and security to the town, and encouraged migration into the area. At present, a number of nationalities and ethnic groups from all corners of the country live together in Butajira, making it a multi-national town.



Butajira market: pepper trading

Butajira was occupied by the Italian army in April 1936. The Italians took control of the town, abolished existing rules and replaced them with their own rule. During this time, new traders including Arabs came into the town. They brought goods like fabricated textiles, salt, sugar, kerosene, needles and blades into the markets. At the same time the number of local bars (*tej bet*, *tella bet* and *arake bet*) increased. The Maria Theresa thaler was the main currency at that time.

The Italians also introduced the construction of modern residential houses in the town. Those houses were used later as the government offices after the Italians left. The Italians also improved the road from Addis Ababa and cleared it further south. They constructed a new road between Butajira and Zway. The Italians left the town defeated in April 1941. Following liberation, the growth of the town accelerated when Butajira became the capital of the Gurage and Mareko *awraja* (province) in 1954. The municipality of Butajira town was established in 1945 by proclamation of the Government of Ethiopia. This proclamation gave 133 towns in the country legal status.

After Butajira became the capital town of the district in 1954, public services like schools, clinics and hotels started to flourish. The town was surveyed for the first time in 1958 by experts from the Ministry of the Interior with the aim of expanding its territory. The expansion failed because of the unwillingness of the

landlords occupying the area. Because of that the governor changed his seat to Adami-Tulu town on the Awassa road in 1960, then to Zway in 1962, as a result of which Butajira lost its status of being provincial capital. Butajira became the capital of Meskan and Mareko District under a new province called Haikoch and Butajira. The size of Butajira town did not increase further until after the 1974 popular revolution in the country [4].

A master plan for the town was prepared in 1967. Based on this plan, the municipality constructed the first roads in the town in 1972. The number of houses registered by the municipality in 1956 was 1,616. 976 of those houses were made of wood and mud, with corrugated iron roofs. The remaining were traditional round huts. Later in 1967, a survey revealed that about 70% of the houses in the town were in bad hygienic condition.

The population, according to some official reports, increased from 3,451 in 1962 to 6,600 in 1971. These statistics also showed that there was a higher proportion of females than males in the town.

2.5 Social services

Education was started in the town early in the twentieth century by religious groups in the mosque and churches, with the establishment of a Quranic school and *Yeques Timihirt Bet* (church school), for Moslem and Christian inhabitants respectively. The first modern school was established in town in 1947 with two teachers in

private premises. In 1954 the school was giving education up to grade three. It was upgraded to teach up to grade 5 in 1955 and then to grade 8 in 1956. In 1957, the students numbered 200, with eight teachers. The school expanded to accommodate more students in 1964 with a female:male student ratio of 1:6. In 1967, the school was named Ras Desta Damtew Junior Secondary School.

At present, there is one technical school, one high school, 14 junior secondary schools and 14 primary schools in the district. The total student number is estimated to be 14,738, with a female to male ratio of 1:3.5. About 77% of the district population are illiterate and less than 2% of the population have education beyond junior secondary school.

Medical services were started by European missionaries that lived in Butajira between 1957 and 1964. These missionaries came with the Sudan Interior Mission (SIM) on a temporary basis. The Health Centre in the town was established in 1976-7. On establishment, the Health Centre was staffed by three health workers: a health officer, a nurse and a junior health assistant. In the first 3-4 years of its work the Health Centre was limited to providing only curative health services. The number of staff and the services rendered were progressively expanded to meet public demand [5]. Currently it provides an integrated curative and preventive health service and has a total of 31 staff, comprising 26 health personnel and 5 administrators. The annual budget of the Health Centre is now 540,000 Birr (approximately US\$ 8,000), of which about 76% is accounted for by salaries.



Health centre in Butajira

At present, in addition to Butajira Health Centre, there are two government health stations, 11 private clinics (5 of which are in Butajira town), 11 functional health posts (5 of which are in BRHP surveillance areas) and 8 private drug vendors (5 in Butajira town) in the District, together serving some quarter of a million people. Four new health posts are also under construction.

Butajira Health Centre has no capacity to manage surgical and obstetric emergencies. The nearest referral centre is 100 km south of Butajira town, at Hosanna. There is no efficient transport facility to the referral centre and the road is often in bad condition. To overcome these

shortcomings, the population of the District and the surroundings, in collaboration with non-governmental organisations (NGOs) have completed the construction of a hospital in Butajira town, although it is not yet equipped and staffed. When the hospital starts to function, it is intended to provide a wider range of medical services that are badly needed in the area.

The rivers around Butajira town were the main source of water until 1980. At that time the municipality after very hard struggles through the years managed to pipe water to the town population from two wells. The source wells for this tap water are tested twice a year and the distribution tanks are cleaned regularly.



The new hospital in Butajira town under construction

2.6 Economy

The backbone of the economy in the district is the agricultural sector. Pepper, coffee and *khat* are the main cash crops produced in the area. *Teff* is also mainly produced for sale. In town most people are engaged in commercial activities. Shops, food and drink establishments, carpentry, and flour mills dominate the town's commercial activities. These are also the main source of revenue for the town municipality.

2.7 Communication and power

The all-weather road that connects Butajira town with Addis Ababa was built in 1966. The town is also connected to the Rift Valley town of Zway by an all-weather road. The villages in the district with few exceptions are connected to the town only by dry-weather roads.

The completion of the all-weather road to Addis Ababa heralded the start of postal service in town. The post office serves five other surrounding districts. It has a capacity of 150 private boxes. A telephone service was started in the 1970s. Currently an automatic dialling service is provided within the town, but outside calls are made through the operator. Plans are being carried out to provide a fully automatic digital service to the town. There are currently 186 customers for telecommunication services, which do not extend outside the urban area.

An electricity supply was started for the first time by an Arab merchant in 1956 from a diesel generator. The

supply of electricity to the town continued to be supplied by a group of Arab merchants until the 1974 revolution. Currently a 24 hour service is available from the Koka hydroelectric power plant.

3. PLANNING AND ORGANISING THE FIELD WORK

3.1 The project design

The objectives of the BRHP, when initiated in 1986, were to generate health-related information, to establish and maintain an epidemiologic research laboratory, to build and strengthen research capability and to develop local capacity in the prevention and control of disease.

The Study district is administratively located in Gurage zone, Southern Nations, Nationalities and Peoples Regional State (SNNPRG). The District is divided for administrative purposes into a total of 82 Peasants' Associations (PAs) in the rural areas and 4 Urban Dwellers Associations (UDAs) in Butajira town. The study has been conducted in a set of nine randomly selected PAs (using the probability proportionate to size technique) and one UDA. In line with its objectives, the initial tasks for the BRHP during 1986-87 were to perform a census of the population in the selected villages to obtain the baseline population and to establish a system of demographic surveillance with continuous registration of vital and migratory events at a household level (Appendix IV). Events registered by the BRHP are birth, death, marriage, new household, out-

migration, in-migration and internal move (migration within the BRHP surveillance villages).

Data are collected monthly by visiting each household. Each household is identified by a unique number within its village, and each individual within their household.

Individuals can thus be characterised by a unique identity number within the entire surveillance system. This unique identity number is ten characters long and is composed of the village number, the house-hold

number and the individual number within the household. Once an individual is allocated a unique identity number in this way, it is permanently associated with that individual, even if they move to a new location (the details of which are separately recorded).

Any adult of the household above the age of 15 years is eligible to respond to the monthly household event collection interviews. The status of each individual is checked during the monthly visit and recorded on a demographic surveillance sheet. Basic demographic, social, housing conditions and health care utilisation characteristics are recorded for each household at entry into the surveillance system and during any re-census process.



Household included in the demographic surveillance

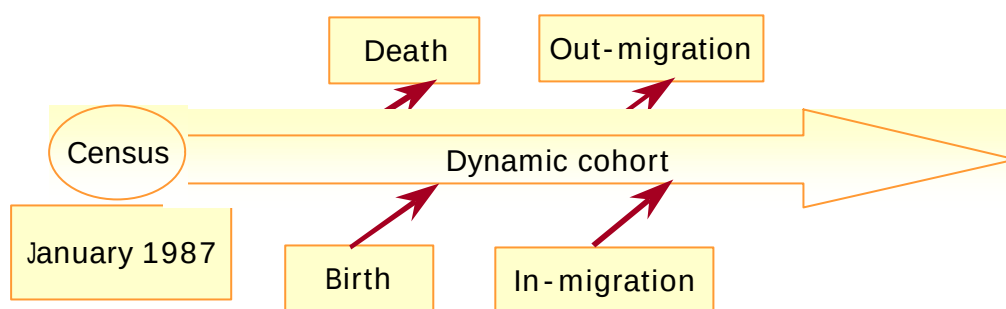


Figure 1: **Study design – prospective surveillance of vital events.**

The surveillance system operates on an open cohort system and is dynamic. Individuals enter and leave the system any time. Since individual movements are tracked regularly, an individual's overall time contribution to the study base can easily be calculated. The individual person-times calculated can then provide denominators for calculation of, for example, mortality and fertility rates within the study base (Figure 1).

Each vital event is registered on a separate form at the household level. Each form contains several questions that are pertinent to adequately characterise the event. The household interviews are conducted by village-based enumerators (enumerators who themselves live in the study villages). Village based enumerators have had at least 10 years of formal education. They were trained at recruitment on the use of the surveillance forms, on how to conduct interviews and on data management in the field. In addition, they receive periodic refresher training on the same

subjects. Job descriptions and interview guides are provided to the field staff to clearly indicate their duties and responsibilities. Data collection is usually undertaken in the morning (0700-1300). Each village is divided into four zones for the sake of simplicity. The enumerator in the village will spend one week in each zone and is not allowed to continue onto the next zone before the allotted time even on completing one zone. This is to ensure that each household is visited at monthly intervals. Completed questionnaires are submitted by the enumerators to their field supervisors on a weekly basis.

Data quality assurance mechanisms have been instituted at several points. The most critical of these is the field supervision. Field supervisors (4 persons) each designated to 2-3 villages and a project coordinator at Butajira perform the immediate supervision of data procedures on a daily basis. These people have served the project from its inception and are vital organs of

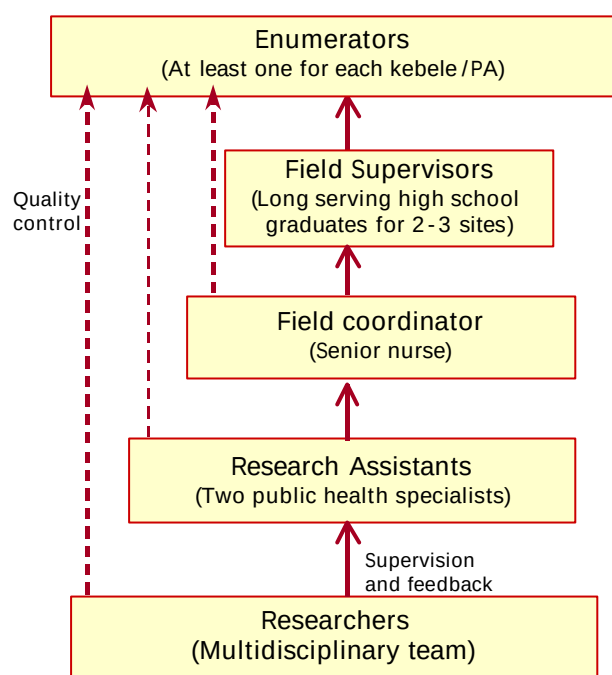


Figure 2: **Scheme of supervision.**

the surveillance system. Their tasks include checking of each and every completed surveillance form and visiting a randomly selected 5% of households each month on a weekly distributed time table. The next level of supervision is performed by the BRHP research assistants (1-2 persons). Research assistants have had public health training at Masters' level. They are responsible for the overall supervision of the surveillance system from the household level to the computer system. They also perform actual data checking at a field level and randomly check on some households every fifteen days.

The highest level of supervision is performed by the researchers themselves (Figure 2).

Data entry is performed at the Department of Community Health in Addis Ababa. Specially developed computer software, using the dBase IV platform, is used to facilitate

automatic data checking at entry. Each event form is entered in a separate transaction file by data entry clerks. The transfer of data from the transaction files to the main database is done by the researchers using a monthly updating function. Data records rejected by the computer for various reasons are returned to the field supervisors for appropriate measures through the research assistant (Figure 3). The introduction of this computer software has substantially improved data quality and management. Efficient management of such a dynamic study base was not possible to achieve with the use of commercially available software. Ideally the data processing should be done at the field level but logistic reasons have precluded the implementation of this so far. Unreliability in the electricity supply, lack of appropriately trained computer personnel and unreliable communication systems (telephone and transport) in Butajira were the main reasons keeping the data processing away from Butajira.

The overall field work planning is the responsibility of the Department of Community Health. Field plans are made by a management team composed of the Ethiopian researchers and research assistants chaired by the Head of Department. Administrative matters and required logistics are arranged through the

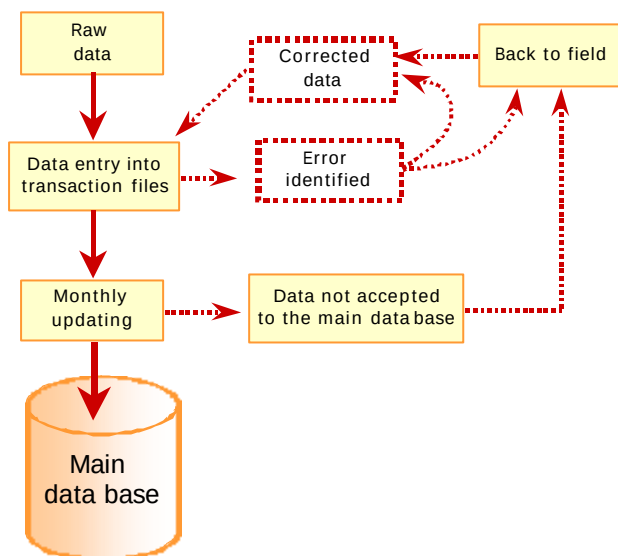


Figure 3: **Data system.**

Ababa and the field co-ordinator in Butajira. At the field level feedback is given to enumerators and field supervisors on an individual basis weekly or as otherwise appropriate. Short meetings are held monthly when everybody gathers to collect their monthly salaries. Overall feedback for field staff is given during these quarterly meetings, which all field staff are required to attend. That also gives them the opportunity to learn from each others' experiences.

3.2 Organisational structure

The Butajira Board, composed of all Ethiopian researchers from all projects operating in the study base and a representative from the local health authority, meet occasionally. The Board is meant to oversee the overall co-ordination of research projects and to maintain ethical standards. This may be seen as a safety valve to avoid duplication of efforts and violation of the rights of the study population.

On scientific grounds periodic censuses are necessary to check on the validity of the surveillance system and to detect any missed events. A second census was done in November 1995 in all study villages. Since the interval between the two censuses was felt to be too long, a third census took place in November 1999. The extent of similarity between the census and the surveillance database illustrates the quality of the continuous registration system.

The BRHP field office in Butajira provides a very good working environment. It was constructed in 1995. Before that the project was housed in a very small room shared with the local Kebele 04 administration in Butajira.

The field compound located behind Butajira Health Centre has offices, computing equipment and guest rooms. The facilities are sufficient to carry out the surveillance activities and potentially to establish a data management centre closer to the field.

Though not yet extensively delivered, some medical benefits have accrued to the populations in the study villages. Many of the surveys conducted within the study base have offered treatment for locally endemic diseases at the time of the interview. Intervention studies such as the ARI study and the indoor air



The BRHP field office

pollution study (see below) have provided complete treatment for the specific diseases targeted in the study.

These projects support the establishment and maintenance of health posts in all rural villages and the training of community health workers. A mental health study (one of the affiliated research projects) established a psychiatric clinic at Butajira Health Centre and assisted in the training of psychiatric nurses.

The study base is now well established and is being utilised for other more focused studies on essential health problems of the country using quantitative as well as qualitative research methods. So far research in the area of childhood respiratory illnesses, reproductive health, mental health and other infectious diseases have been conducted utilising the study base infrastructure.

Many of these projects have been part of doctoral studies for Ethiopian candidates as well as for Swedish partners.

3.3 The training

BRHP has contributed to human resource development and the building of research capacity at the Faculty of Medicine, Addis Ababa University. Nine doctoral study projects have been carried out in the study base. So far, four have been completed and defended. The remaining five are ongoing projects to be completed in the coming 1-2 years. These doctoral studies are mainly registered and supervised by Umeå University and other Swedish Institutions. Three of the nine doctoral studies are for Swedish researchers. The training is conducted in a sandwich model that allows doctoral researchers to stay close to

their mother institution and carry on their routine teaching and other responsibilities. The sandwich model of training also significantly reduces the risk of the brain drain that may occur when people are sent for long-term training abroad. The Department of Community Health has witnessed that with only one out of four staff returning from long-term studies in North America.

Research capacity was also enhanced through the short-term training programme offered by Umeå University. Seven academic staff of the Faculty of Medicine have attended the summer course on field research methods in Umeå. The Department of Community Health also offered a research methodology course for public health workers in Ethiopia. Over ten health professionals representing all but two Regional Health Bureaux in Ethiopia have attended a research methodology course in Nazareth, Ethiopia.

This training is one of the direct outputs of the BRHP in both collaborating institutions. It is also an incentive for collaborating Ethiopian researchers to receive further and additional training on recent advances in research methodology.

3.4 The collaboration

The BRHP is a collaborative research undertaking between the Department of Community Health, Faculty of Medicine, Addis Ababa University in Ethiopia, and the Department of Epidemiology and Public Health (now Department of Public Health and Clinical Medicine), University of Umeå, Sweden. The collaboration is based on mutual understanding and interest on both sides.

The collaboration started as an individual doctoral study project. As envisaged from the beginning, it grew into departmental collaboration along with the development of the study base infrastructure.



Part of the BRHP team at the computer in the field

The intensity and diversity of the research activities also necessitated a wider participation of multidisciplinary researchers. At present, the Butajira study base has proved that it is in a good position for undertaking scientifically rigorous studies on major public health problems in a developing country setting. Thus, the collaboration is expanding its scope to involve more departments and institutes in the collaborating and other countries. This growing collaboration will facilitate the maximum utilisation of the study base and improve on the quality of the research output.

The BRHP study base has required a tremendous amount of personal and institutional investment. The generous and continuous financial support from Sida/SAREC has been very crucial throughout its long life. However, the strong professional commitment of collaborating researchers on both sides has been the major driving force. The Ethiopian Science and Technology Commission and Addis Ababa University, particularly the Faculty of Medicine, have nurtured the collaboration over the entire period of the base's existence.

4. BASIC DEMOGRAPHY

4.1 The baseline population

The initial aim of the Butajira project was to study disease occurrence in terms of mortality. It is thus natural to start by describing the baseline population as of 1st of January 1987, the starting date for the documentation of cases of death, migration and person-time of follow-up.

Table 1 shows the description of the population of 28,616 individuals enumerated during the census and present at the beginning of 1987 by age, sex and area of residence. The stratification of this analysis is motivated by different living conditions in the highland, lowland and urban areas.

The age structure of the population is further illustrated in Figure 4. It reflects a surplus of females, especially in reproductive ages, through a male:female sex ratio of 0.93.

Table 1: **Description of baseline population in Butajira by age, sex and area.**

Age	Urban		Highland		Lowland		Total		Overall
	Male	Female	Male	Female	Male	Female	Male	Female	
0	46	55	117	133	98	104	261	292	553
1-4	296	307	841	877	705	718	1,842	1,902	3,744
5-14	739	795	2,254	2,261	1,661	1,697	4,654	4,753	9,407
15-44	684	802	2,623	3,051	1,915	2,254	5,222	6,107	11,329
45-64	197	252	717	688	495	403	1,409	1,343	2,752
65-84	78	128	212	177	95	97	385	402	787
85-	6	4	14	7	8	5	28	16	44
Total	2,046	2,343	6,778	7,194	4,977	5,278	13,801	14,815	28,616

Of all women, 41% are in 15-44 year age group. The broad-based distribution includes 48% children under fifteen and 15% under-fives. Three percent of the population are 65 and above. The distribution may be characterised through a *dependency ratio* of 1.03, implying that there are equally many outside ("dependants") as within the 15-64 age group ("supporters").

These indicators may also be used to compare the different peasants' associations (PAs) within the three areas (Table 2).

Table 2 shows that there is a surplus of females in all the study populations, particularly so in Butajira town. The urban area also dis-

plays the highest dependency ratio, reflecting an ageing population and possibly also the absenteeism of men in productive ages. The proportion of women in reproductive ages is also lowest in the town.

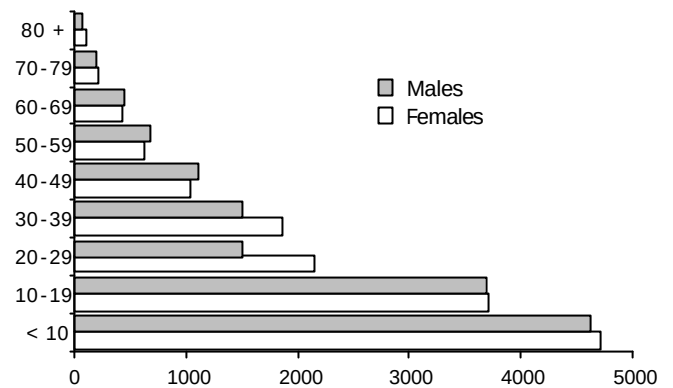


Figure 4: **Structure of the baseline population in 1987.**



A family in one the study sites in front of their house

Table 2: **Basic demographic indicators of the ten study populations.**

Area	PA population	Sex ratio	Depend- ency ratio	Women of reproduc- tive age	Proportion of children under 15	Proportion of children under five
Urban	Butajira 4 (4,389)	0.87	1.27	0.34	0.51	0.16
	Mmeskan (2,769)	0.95	0.96	0.43	0.46	0.15
	Bido (1,824)	0.92	1.05	0.39	0.46	0.14
Highland	Dirama (2,073)	0.96	1.07	0.42	0.50	0.15
	Yeteker (3,634)	0.92	0.99	0.41	0.46	0.14
	Wrib (3,663)	0.96	0.89	0.46	0.45	0.13
	Mjarda (1,802)	0.94	1.12	0.40	0.51	0.15
	Bati (2,835)	0.92	1.09	0.40	0.50	0.15
Lowland	Dobena (3,257)	0.96	0.96	0.45	0.47	0.16
	Hobe (2,361)	0.95	0.98	0.44	0.48	0.17

Table 3: **Basic social, environmental and cultural characteristics of the baseline population.**

Characteristic	Category	Urban %	Highland %	Lowland %	n
religion	Moslem	64.7	81.5	72.7	20,417
	Christian	35.3	18.5	27.3	6,532
ethnicity	Sodo	6.6	15.7	2.3	2,367
	Dobi	12.5	4.0	0.7	983
	Meskan	71.3	63.7	33.8	13,225
	Mareko	0.0	0.5	32.4	2,853
	Silti	9.6	16.2	30.9	4,993
literacy*	literate	52.3	10.7	19.9	3,505
	read only	6.8	4.4	9.3	1,131
	illiterate	40.9	84.8	70.9	12,857
household literacy	literate adults	79.5	23.1	36.4	10,027
	none	20.5	76.9	63.6	17,480
literacy comparison between head and spouse	both literate	36.6	6.3	14.3	2,620
	head only	33.2	14.4	21.4	3,865
	spouse only	5.8	1.2	1.2	359
	neither	24.4	78.1	63.1	13,073
household members	mean	5.69	5.21	5.22	
	s.d.	2.60	2.34	2.27	
healthcare source	health system	92.7	93.6	19.3	17,225
	pharmacy	4.6	3.2	75.9	7,569
	traditional	1.4	2.2	3.6	664
	self-treatment	1.3	1.0	1.3	298
source of water	river	10.7	80.4	22.0	12,095
	well	2.7	19.2	78.0	9,352
	urban supply	86.6	0.3	0	3,239
roof of house	thatched	9.5	95.2	95.7	20,621
	corrugated iron	90.5	4.8	4.3	4,310
livestock in house	yes	13.8	90.1	87.8	19,234
	no	86.2	9.9	12.2	5,466

* 9,218 individuals (32.2%) were too young to be literate

The aim of the census was to enumerate the study population (i.e. to get a baseline for the continuous surveillance) and also to get some background information about the population. Information collection was based on a structured questionnaire containing questions concerning basic demographic characteristics, cultural patterns, household structure, environmental conditions, health care utilisation and recall of vital events.

Table 3 compares the three areas in terms of these basic characteristics. It shows that this is a predominantly Muslim community, especially in the rural area. There is ethnic variety, although there is one major ethnic group in the urban and highland areas. Illiteracy is very high, with two out of three house-holds in the rural area having no literate adults in the household while, in the town, 80% of the households have at least one literate household member. As expected, there is inequity in the household with regard to literacy, with mainly the head of the household being literate. Urban households tend to be somewhat larger than rural families.

Socio-economic conditions are illustrated from water supply, house roofing and the presence of livestock in the house. While there is a piped water system in the town, highland households mainly utilise river water and lowland households have wells. Typically the houses in the town have corrugated iron roofs and the *tukuls* in the rural areas have thatched roofs. The latter usually goes with keeping livestock in the house.

4.2 Population development 1987-96

After the census was completed, demographic surveillance with monthly visits to individual houses was introduced. The status of each and every member of all households was checked and recorded on the day of the visit on a demographic surveillance sheet. Information collected included changes that had occurred since the previous visit - births, deaths, causes of death and migration. This implies that, at any point in time, the current ("active") population can be defined. Thus, during the ten-year period, the study population was observed to increase from a baseline population of 28,616 to 37,323 at the beginning of 1997, implying a mean yearly growth rate of 2.7%. The cumulative population growth of the three study areas is shown in Figure 5. Growth rates were consistently lower in the highland PAs (Table 4).

As indicated in Table 4, the major difference in population growth between areas is explained by urbanisation, with a marked excess of births to deaths and net in-migration in the urban area. During ten years of surveillance a total of 5,143 deaths and 15,667 births were registered in the area, generating the pattern of vital statistics by area as shown.

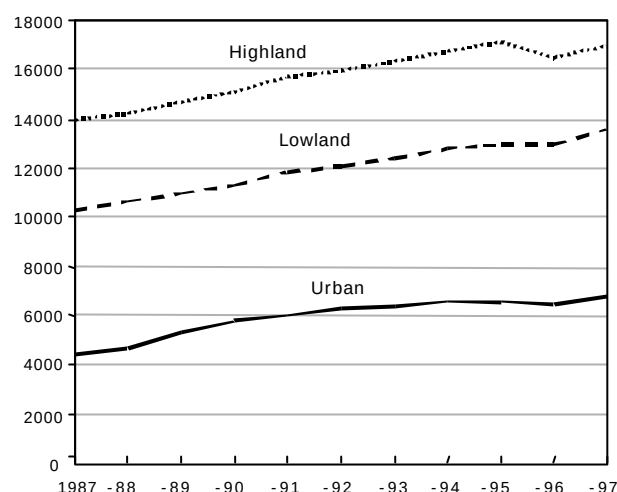


Figure 5: **Population development in the three study areas 1987-96.**

4.3 Overall mortality pattern

During the ten years of surveillance, a total of 336,074 person-years of follow-up were generated. Thus, by relating the observed total number of deaths to this study base, we get a *crude mortality rate* of $5,143/336,076 = 15.30$ per 1,000 person-years. Based on the Poisson distribution, the 95% confidence interval for this estimate is 14.9 to 15.7.

From Table 5 we may conclude that there are major differences in mortality between areas and consistently higher mortality among males.

To provide information about age-specific mortality patterns, Table 6 shows the overall infant, childhood and adult mortality rates by area and sex.

A sex-specific graphical illustration of mortality patterns in the three areas over age is shown in Figure 6.

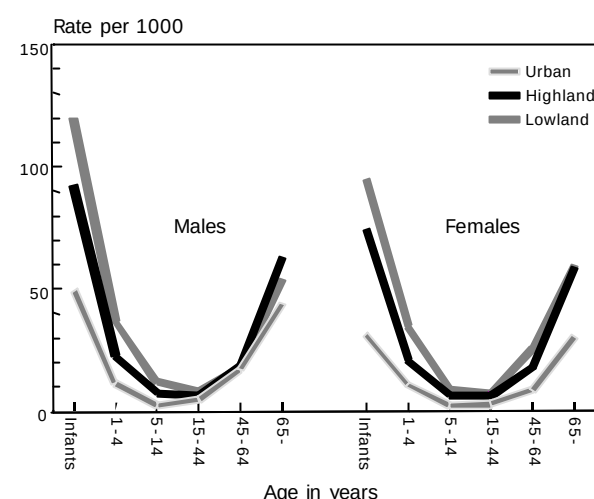


Figure 6: **Sex-specific mortality rates by area.**

These "bath-tub" curves show the higher mortality in the lowlands for both males and females.

An interesting pattern now emerges where the excess male to female mortality still holds in the urban area for all age groups while, in the lowlands, there is a shift to the disadvantage of females above 45 years of age.

Table 4: **Patterns of fertility, mortality and growth by area 1987-96.**

	Urban	Highland	Lowland
Population growth rate	4.46%	1.96%	2.85%
births	2,448	7,007	6,212
deaths	468	2,330	2,345
birth:death ratio	5.23	3.01	2.65
people immigrated	4,764	5,992	5,484
people outmigrated	4,290	7,672	6,022
in:out migration ratio	1.11	0.78	0.91

Table 5: **Crude mortality rates in the Butajira study base per 1,000 person-years by sex and area.**

Area	Sex	Crude mortality rate per 1,000 person- years	95% CI
Urban	M	9.26	8.13 to 10.39
	F	6.70	5.79 to 7.61
Highland	M	16.00	15.10 to 16.89
	F	13.69	12.88 to 14.50
Lowland	M	20.93	19.76 to 22.10
	F	18.32	17.25 to 19.39
Total	M	16.61	15.98 to 17.24
	F	14.07	13.51 to 14.63
Overall	M & F	15.30	14.88 to 15.72

Table 6: **Mortality rates per 1,000 person-years stratified by age, sex and area.**

Age	Urban		Highland		Lowland		Total		Overa ll
	Male	Female	Male	Female	Male	Female	Male	Female	
0	49.4	31.6	92.5	74.5	119.8	95.0	96.0	75.4	85.7
1 - 4	11.2	10.9	22.5	20.9	36.6	34.9	25.7	24.3	25.0
5 - 14	2.69	2.50	7.57	6.55	11.9	9.25	8.22	6.73	7.46
15 - 44	5.00	2.81	6.60	6.57	8.04	7.38	6.85	6.23	6.51
45 - 64	17.1	8.59	18.8	18.1	18.2	25.6	18.4	18.6	18.5
65 -	43.9	30.4	63.2	58.6	54.1	59.8	57.0	50.2	53.6
Total	9.26	6.70	16.00	13.69	20.93	18.32	16.61	14.07	15.30

The highland area seems to be in an intermediate position with regard to this "transition" (Figure 7).

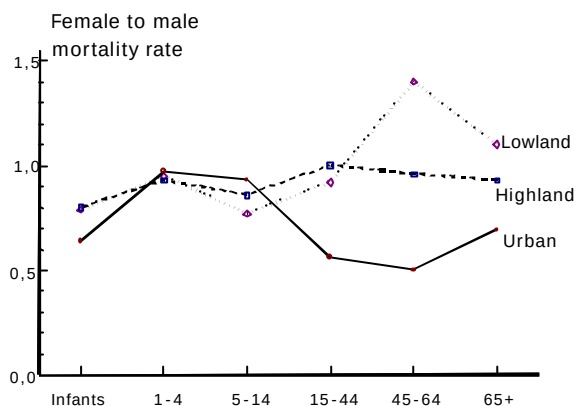


Figure 7: **Comparing life-span mortality between sexes by means of female to male mortality ratios for the three areas.**

The rural to urban mortality differentials are somewhat more pronounced for females, especially for adults.

Figure 8, again, illustrates the disadvantageous situation of rural women while, for men, there is only a rather small excess mortality in rural areas. The most marked area differences in mortality occur among children.

The 2,456 under-five deaths represent nearly 50% of all deaths.

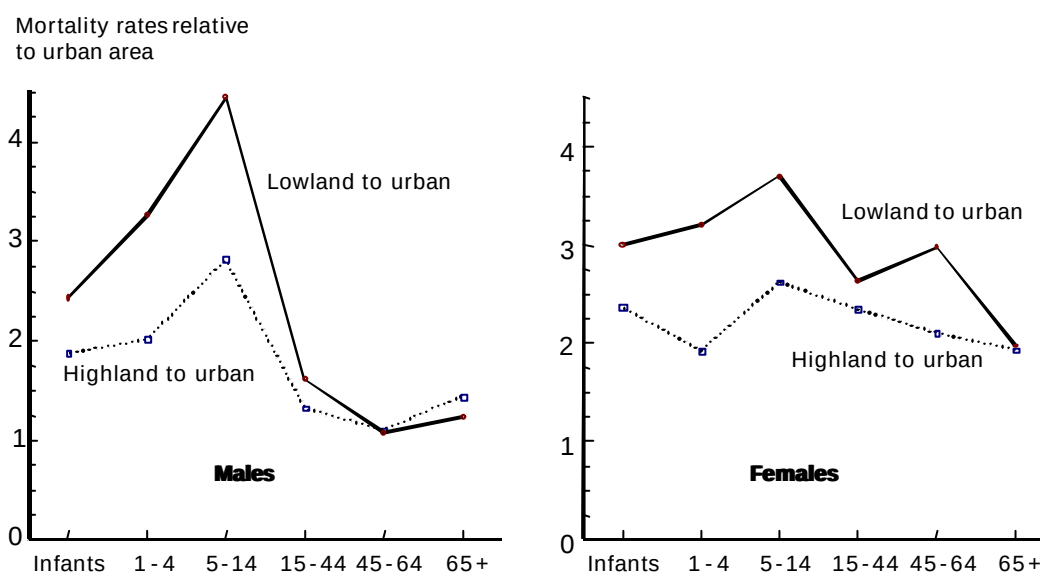


Figure 8: **Comparing life-span mortality between areas for males and females.**

Half of the under-five deaths occurred during the first year of life, half of which fell below 2 months of age. From the age-specific mortality rates we can estimate the cumulative mortality throughout life. Thus, among live births 4.2% are estimated to die during the first two months of life, 8.0% before one year, 16.6% before 5 years, 36% before 15 years, and 56% before 65 years. There were substantial variations between areas also with regard to under-five mortality ranging from 80 per 1,000 person-years in the urban area to 219 per 1,000 person-years in the lowlands (Figure 9).

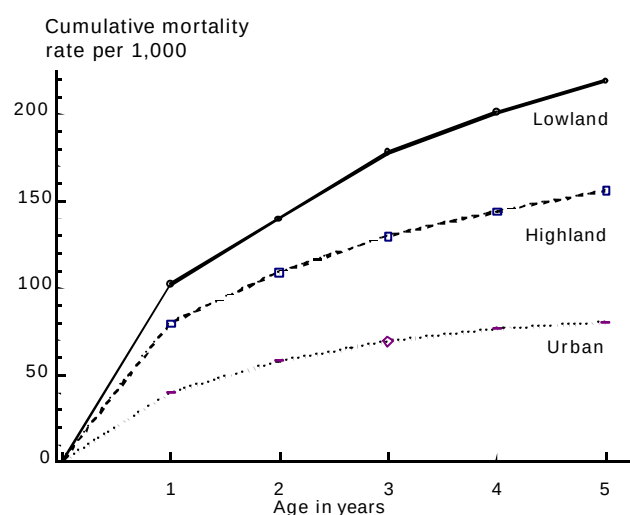


Figure 9: **Cumulative under-five mortality rate by area in Butajira during 1987-96.**

From the age-specific mortality rates a life expectancy at birth of 50.8 years is estimated, 49.3 years for males and 52.3 years for females.

Figure 10 shows gender differences in life expectancy at birth by area and Figure 11 the expected remaining longevity by sex from different ages.

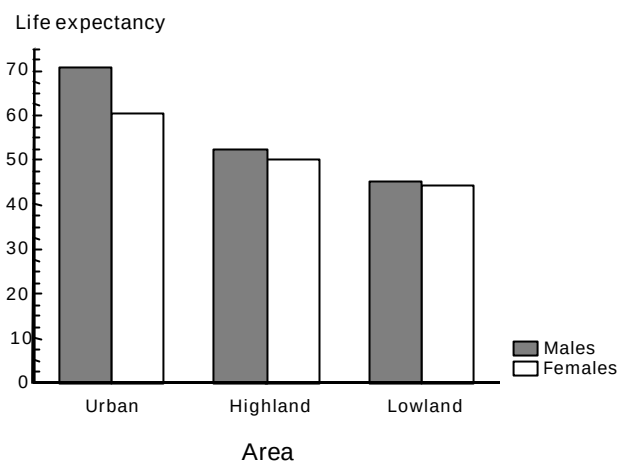


Figure 10: **Life expectancy at birth by sex and area.**

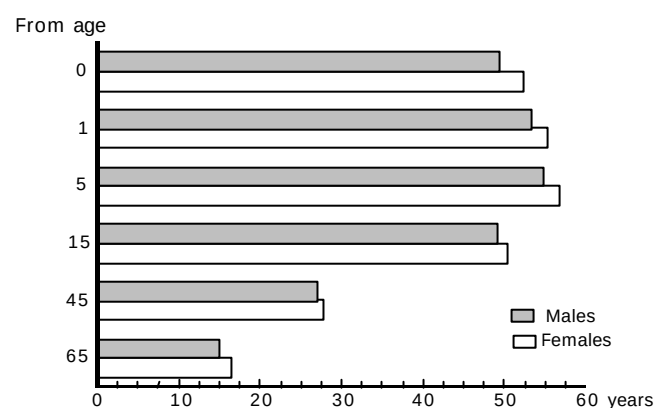


Figure 11: **Remaining life expectancy from different ages by sex.**

Table 7 summarises the mortality pattern for the ten study populations in terms of the age-specific mortality rates and life expectancy consequences from different ages.

Table 7: **Mortality characteristics of each of the ten different study populations.**

Area	PA	Age-specific mortality rates/1,000 person-years						All ages	Life expectancy at birth
		0	1 - 4	5 - 14	15 - 44	45 - 64	65 +		
Urban	Butajira	40.5	11.1	2.59	3.83	12.3	35.4	7.91	69.2
	Mmeskan	71.8	18.8	6.23	6.52	18.6	67.6	13.8	55.1
Highland	Bido	97.8	27.2	11.20	9.52	17.9	55.2	19.9	48.4
	Dirama	96.6	29.3	8.75	7.31	15.5	91.2	16.6	49.8
	Yeteker	64.9	19.8	6.76	6.22	17.4	58.7	13.5	56.2
	Wrib	95.5	19.6	5.23	5.46	21.2	51.7	13.8	55.4
Lowland	Mjarda	100.3	32.8	10.10	9.12	21.6	54.7	18.4	47.2
	Bati	114.4	37.5	11.50	9.28	22.2	51.1	20.6	45.2
	Dobena	128.2	42.4	11.10	7.40	22.8	64.9	21.8	44.4
	Hobe	83.2	28.4	9.23	5.24	18.5	54.2	16.5	53.2
Total	All PAs	85.7	25.0	7.46	6.51	18.5	53.6	15.3	50.8

5. APPLYING THE STUDY BASE TO SPECIFIC STUDIES

5.1 Acute respiratory infections among children

From various ARI studies, several reports have been completed and a new study is at the planning and field staff recruitment stage. All of these studies have been nested in the demographic surveillance system, using its output - the population database - as a sampling frame for identifying specific study populations and utilising the monthly routine surveillance to follow the study subjects. The existing infrastructure of the BRHP field organisation has been used for co-ordination of project activities, supplemented by extra staff who have been specially trained for the data

collection activities of each specific project. This also has meant that over the years a pool of trained field staff, enumerators, has grown to become a resource for new activities.

A variety of study designs, focussing on different aspects of ARI epidemiology, have been used to give a broader view of the problem area, as shown in Figure 12. After an initial cohort study using the routine surveillance to follow the population, there was a cross-sectional prevalence study, a nested case-referent study, a cross-sectional survey, a prospective cohort study and a step-wedge design intervention study. The new study is to be a cohort and nested case-referent study.

Cross-sectional	Cohort	Case-referent
Prevalence study of potential respiratory disease bacteria in children under five	Initial study of cause of death pattern in under-fives	Nested study of determinants of under-five mortality
Mothers' perceptions of illness and practices in care of children with ARI	Prospective study of morbidity patterns in under-fives	Nested study of ALRI and indoor air pollution
	Step-wedge design case management intervention	
	Prospective exposure assessment for indoor air pollution	

Figure 12: **Study designs and different components of studies of acute respiratory infections.**

The ultimate objective of the ARI studies has been the design of an effective and appropriate case management protocol for pneumonia in children under 5 years of age at the community level. Existing data suggested that up to 35% of infant and child mortality in Ethiopia is associated with pneumonia. Thus the ARI component of the BRHP aimed at identifying major deficiencies and obstacles in health care delivery, reviewing mothers' perceptions and practices in the care of children with ARI and characterising aetiologic agents and risk factors. A carriage study showed that 85% of children in the project area carried *Haemophilus influenzae*, 83% *Moraxella catarrhalis* and 90% *Streptococcus pneumoniae* in the nasopharynx.

5.1.1 Mortality

An early application of the BRHP study base grew out of the findings from the verbal autopsy data of the first two years (1987-88) of the demographic surveillance. As in many developing country settings, acute respiratory infections (ARI) were found to constitute the major identifiable cause of death among infants and children up to five years of age, exceeding the mortality from diarrhoeal diseases in all the rural areas in the study base [6]. There were also shown marked differences in both ARI mortality and all-causes mortality among under-fives between different parts of the study area, highest in the rural lowland and lowest in the urban sub-population of the study base.

ARI mortality was particularly affected by parental factors, including illiteracy of the father, and environmental factors, including lack of windows in the house, and less so by breast-feeding, milk, and protein supplementation, which instead significantly affected diarrhoeal mortality [7].

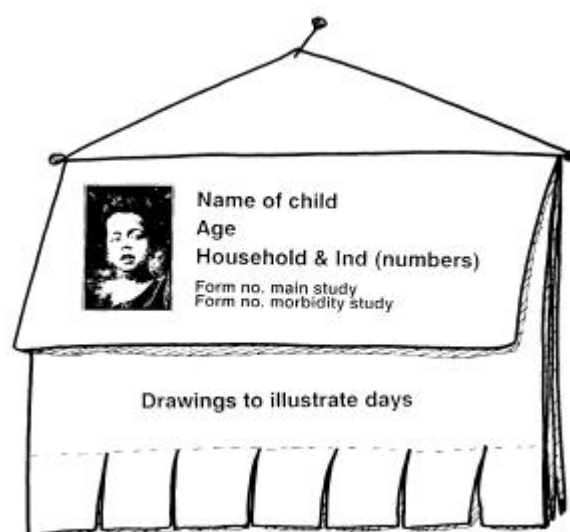
5.1.2 Morbidity

The prospective morbidity study was based on weekly home-visits. A cohort of 1,315 children under the age of 5 years were followed weekly, including the use of a recall card, over a one-year period. The recall card was a simple calendar with a photograph of the child, on which mothers recorded perceived illness on a daily basis. The weekly interviews collected this information as well as limited clinical observations made by the field workers, who were specifically trained for this. The results of the daily reported morbidity were assembled and used to define episodes of disease entities such as ARI, diarrhoeal disease and febrile illness. Incidence of disease episodes were then related to individual risk factors in a multivariate Poisson regression model.

5.1.3 Intervention

Intervention strategies to reduce ARI mortality and morbidity were reviewed. The implementation of an ARI intervention package consisting of health posts, community health agents and a handful of inexpensive

drugs to reduce mortality from ARI was a logical follow up. Such a package is affordable and sustainable in a poor country like Ethiopia. Before launching the intervention, it was important to talk to leaders of communities on the aims, processes and advantages including responsibilities of the communities. The impact of the intervention can be evaluated almost continuously by the monthly surveillance in terms of under-five or infant mortality rates or cause-specific mortality rates comparing intervention areas with previous years and with control areas.



5.2 Women's health

5.2.1 Maternal mortality

Maternal mortality is difficult to estimate from rural populations in low-income countries due to the relative rarity of the event and lack of official birth and death registration. Hospitals are scarce and means of reaching hospital from rural areas are difficult or impossible. Several studies from Ethiopia have reported quite

different maternal mortality rates. These studies present maternal mortality from different perspectives; some are based on hospital data while others are based on community based data.

Maternal mortality ratio (MMR) relates the number of maternal death cases to the number of live births in the same period. A maternal death is defined as the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and the site of the pregnancy, from any cause related to or aggravated by the pregnancy or its management but not from accidental or incidental causes.

The BRHP is a framework for community based studies and, as such, a suitable platform for studies of maternal mortality. However, the total population under study is about 37,000 inhabitants and this is almost too small a sample size for MMR estimation on a yearly basis.

Finding cases of maternal death requires continuous surveillance and verbal autopsy interviews of relatives to the deceased and ideally also pregnancy identification and follow-up. Since no information has been collected about pregnancies there is likely to be an underestimation of maternal deaths occurring during early pregnancy.

Estimation techniques have been applied in the area for reasons of validating and comparing these with the direct estimate from the surveillance data. The *sisterhood method*,

enquiring about adult respondents' sisters ever married and their experience of maternity-related death, is an efficient and cheap method of estimating MMR where no information on mortality and birth is available. The *case-control approach* is an indirect estimation technique where deaths in females aged 15-49 years, identified from the surveillance system, are used as cases. Controls were age-matched women from the same village. With exposure defined as experiencing maternity during the previous 12 month period, the proportion of female deaths attributable to that exposure estimates the number of maternal deaths. Extrapolation to the MMR was made using the surveillance estimate of the total number of live births.

In the ten-year period, 1987-1996, 512 cases of female death in the 15-49 year age group were identified. Applying the different methods for estimation of MMR resulted in estimated maternal mortality ratios in the range of 440 to 665 maternal deaths per 100,000 live births [8].

In comparing the direct surveillance methods with indirect methods, such as the sisterhood method or the case-control method, the discrepancies are due to both missed cases during surveillance and methodological drawbacks of the indirect methods. The sisterhood method gives a proxy for MMR for a period of 20 years due to the method's in-built time lag against a background of secular reductions in general mortality. The case-control

method cannot distinguish between deaths in the 42 days following delivery and later events.

5.2.3 Domestic violence

Physical, psychological and sexual violence against women is a global problem, crossing cultural, geographic, religious, social and economic boundaries. In the last decade domestic violence has been recognised as a public health issue on the international arena, most recently in the UN World Conference on Human Rights (1993), the International Conference on Population and Development and the Fourth Conference on Women in Beijing (1994). Physical violence includes beatings, stabbing, burning, and the intentional deprivation of food. Psychological violence includes the enforcement of strict isolation, constant denigration and public humiliation. Sexual abuse includes sexual harassment and coerced pregnancy. Besides traumatic bodily injuries, domestic violence is also associated with less well-defined somatic complaints, including chronic headaches, abdominal and pelvic pains and muscle aches.

Violence against women has been associated with a number of adverse reproductive health consequences such as unwanted pregnancy, miscarriage, chronic pelvic pain, and STDs/HIV. The child in a family of domestic violence is at risk for child abuse and death and adulthood depression. The increased risk of low birth-weight and intrauterine growth

retardation associated with domestic violence should be interpreted as a consequence of maternal stress seriously threatening the survival of the offspring.

The first study on domestic violence in Ethiopia was performed within the BRHP surveillance site as a cross-sectional survey including 673 married women as a master thesis [9]. A life-time and 3 month prevalence of physical spousal violence of 45% and 9.7% were reported respectively. Occurrence of physical violence showed no statistically significant difference regarding background factors as literacy, ethnicity, urban/rural or religion. Being under the age of twenty was associated with a higher than three months prevalence of domestic violence. Women who were able to negotiate their own marriages were less likely to report physical violence. Women having a history of parental spousal abuse in the family of origin, spousal abuse of alcohol or their own *khat* abuse, reported more physical abuse.

5.3 Mental health

It is gradually being recognised that mental illnesses are public health problems in developing as well as developed countries [10].

A comprehensive cross-sectional survey of mental health has been performed as a doctoral thesis linked to the BRHP [11]. Out of all 12,531 adults in the study base, 10,468 were reached and interviewed according to a self-reporting structured instrument

for mental distress with a participation rate of 86%. Female sex, marital status other than single and illiteracy were factors significantly associated with mental distress. Prevalence among women was 19%. A life-time suicide attempt was reported by 332 adults (3.2%) and of these the majority were females.

In order to institute policies and strategies to deal with mental illnesses, their prevalence in specific communities must be determined. In Ethiopia the priority status accorded to mental health in the National Health Policy has not yet been translated into either strategies or actions to tackle mental illnesses [12]. One important reason is the lack of data on the extent of mental disorders in the general population.

The Butajira Study Base has been utilised for carrying out a number of surveys of adult psychopathology using the Self Reporting Questionnaire (SRQ) and the Composite International Diagnostic Interview (CIDI). Specifically, the prevalence and sociodemographic correlates of mental distress, schizophrenia, affective, anxiety, suicidal behaviours and substance use disorders were studied and reported [13-19]. In addition, a study of mental and behavioural disorders in children was recently completed using the Diagnostic Interview Schedule for Children and Adolescents (DICA). These studies have shown that mental and behavioural disorders are important public health problems among adults and children. Mental

distress as defined by the SRQ was prevalent among 17% of adults. It was found that 3.7% of adults were problem drinkers and 3.2% gave a history of suicide attempts. The prevalence of psychotic disorders and mood disorders, using the CIDI, were 0.6% and 5.0% respectively. The prevalence of substance dependence was 4.1%. The prevalence of mental and behavioural disorders in children, as defined by the DICA, was 3.5%.

A study to describe the course and outcome of schizophrenia and bipolar disorders in Butajira District is currently underway. The study is a collaborative undertaking between the Department of Community Health of Addis Ababa University and Amanuel Psychiatric Hospital of the Ministry of Health of Ethiopia and the Department of Psychiatry of the Umeå University in Sweden. The study is being supported by a grant from the Theodore and Vada Stanley Foundation in the USA.

Preliminary preparations for the study, including development of interview instruments and training of study workers were completed in October 1997. A trial run of the study as pilot project was conducted between November 1997 and February 1998 in a sub-district similar to, but outside, the study area. Data collection for the main study was started in March 1998. The plan envisages the use of the psychoses and affective disorders module of the CIDI and key informants from each of the 44 sub-districts to identify potential cases. This would involve

the application of the CIDI to the entire district population 15-49 years of age (estimated to be over 100,000). A total of over 32,000 adults between the ages of 15 and 49 have so far been interviewed in 27 sub-districts (kebeles).

5.4 Master of Public Health Studies

The Department of Community Health has been running a Master of Public Health programme since 1984. The training consists of coursework, residency attachment programmes and research work. The Butajira study base has provided a continuous forum for research and residency work for these MPH students. Masters' students take research areas that are being developed by researchers in the department to avoid over-burdening the study population.

Besides enhancing the efficient utilisation of resources and personnel in the study base, the Masters' studies have made significant scientific contributions. Areas of Masters' studies include the following broad subjects: child feeding and nutrition, drug utilisation patterns, domestic violence, migration, reproductive health, mental health and health services assessment. To date the BRHP has involved 15 Masters' students from the Department. (Appendix I). The Ethiopian collaborating researchers were able to take heavy loads of Masters' studies supervision since students are coming to an established study base and

working in line with existing research interests.

6. MAJOR FINDINGS SO FAR

6.1 The public health of Butajira 1987-96

The Butajira project has documented high under-five and infant mortality in rural Ethiopian communities. Under-five mortality accounted for 50% of all deaths. ARI contributed one fifth of the under-five mortality and one third of the infant mortality. ARI was more common than diarrhoea as a cause of death among infants.

Based on the first ten years of surveillance cumulative mortality throughout life has been estimated. Thus, among live births 17% are estimated to die before 5 years, 36% before 15 years and 56% before 65 years. The overall life expectancy has been 49.3 years for males and 52.3 years for females.

The Butajira study base was also used to perform a one-year prospective morbidity study based on weekly home-visits. Overall, illness was reported in 5.8% of 11,216 person-years among the under-fives. ARI and acute diarrhoea were the commonest conditions. ARI was responsible for 48% of the under-five illness load and 5.7% of all ARI were acute lower respiratory tract infections. ARI mortality was particularly affected by parental factors, including illiteracy of the father, and environmental factors, including lack of window in the house. Parental

factors, such as illiteracy were also linked to morbidity. In summary, housing factors mainly affected mortality while sanitation, health behaviour and reproductive health factors mainly affected morbidity. Urban/rural residence and parental factors affected both mortality and morbidity.

The extension of health services to the underserved rural population has been an explicit goal in national development plans in Ethiopia over the last decade. The number of health units and health personnel in rural areas and small towns has indeed increased. But has this also meant an improved access to the services? This study illustrates that the outreach of health services within a rural district can be very limited and that large inequalities exist. Over 70% of the population in the small town of Butajira used the health care in connection with reported illness whereas less than 30% of the people outside the town did so. The magnitude of this problem is shown by the fact that the population of peasants is about 20 times that of the town population and that they report 2-5 times more illness. The perceived morbidity in the urban area, the rural highland and the rural lowland could be contrasted with the mortality rates as recorded in the continuous surveillance of vital events during the same year. A high rate of reported illness was associated with high mortality rates, whereas the proportion of clinic users was highest among those with the lowest rate of

reported illness and with the lowest mortality, i.e. in the urban area.

The differences in access to and utilisation of preventive services are even more dramatic. Whereas 70% of mothers in the town used the antenatal services, less than 20% of the mothers in the peasant households did so. The vaccination programmes reach the town children but practically none of the children in the rural areas nearby [1].

It is also obvious that this rural community is a society in transition and urbanisation. Over the ten-year period the overall growth rate has been 2.7% per year on average and as high as 4.5% in Butajira town. The rural to urban mortality differentials are most marked for children, and to the disadvantage of rural women. While, for instance, female mortality 45-54 years in the town is half that among males, it is well above that of males in the rural lowlands.

In summary, this project highlights a gross discrepancy between needs for and access to organised health care at district level, where available health services also seem to work below capacity. Equity in health care is far from having been achieved. No doubt improvements can be made within existing limited resources through better organisation and more attention to outreach programmes and through developing ways to work with the people, empowering them to engage in solving their own health problems and demanding their own share of health resources.

6.2 Methodological experiences

Since the basic *raison d'être* for this type of population based surveillance arises out of a lack of civil registration infrastructure, it follows that appropriate methodologies have to be implemented from first principles. Thus in many ways the BRHP has pioneered methods of epidemiological and demographic surveillance that can be appropriately applied in this kind of setting.

When the Butajira project started in the late 1980s, personal computers were still in early stages of development and there was very little experience of their use in sub-Saharan Africa, particularly in remote locations such as this [20]. Initially therefore it was unrealistic to think of computerisation of the data on-site, and even at Addis Ababa University facilities for handling the data were limited at this stage. Records were originally entered as plain text strings using a simple text editor. Whilst this served a useful function for later analysis, it did not facilitate easy interaction with the data, nor permitted real-time error checking and quality assurance procedures. As well as the basic requirements for processing this kind of data, a special issue arose in the Ethiopian context because of the unique calendar system in use, in which the year starts in mid-September, has 13 months (12 of 30 days and one of 5 or 6) and runs 2,809 days behind the international (Gregorian) system. Since field workers were much more conversant

with the local Ethiopian system, this as used for dates in the field, which were then converted as part of the data entry process. In the initial phases of the project, when data entry was in text strings, this conversion process added further difficulties.

Following a decision in 1993 that a re-census of the entire study population was called for in order to validate and update the study base, new forms for the census itself and continuing vital event surveillance thereafter were developed. These were translated into the common local language, Amharic, and field tested. A sample of the Birth Registration form (English and Amharic versions) are appended. Experience shows that, given the basic educational level of most field staff, the extra time and effort involved in using local languages (and also in this case the local calendar) is well spent in terms of data quality.

Prior to the 1995 re-census, and in the light of developments in computer hardware and software in the meantime, a new software system for data entry and storage was developed using a relational database approach [21]. The basic design of this system is shown in Figure 13.

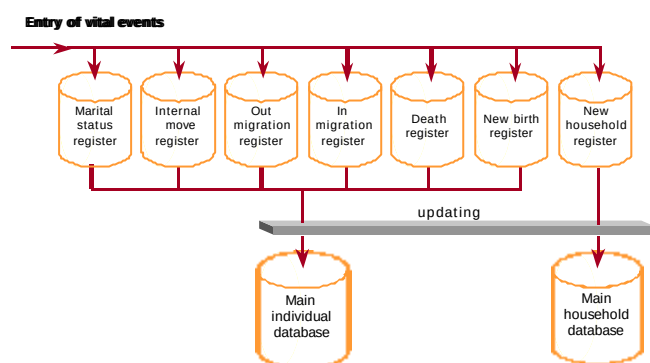


Figure 13: Conceptual design of the BRHP software system.

Within the data checking part of this system, implemented under dBase IV, it was possible to include a routine for automatic conversion of date variables from the Ethiopian format on entry. The development of this new system involved a considerable amount of effort and difficulty, not least as a result of the relative scarcity of computer trained personnel in Ethiopia at the time. Much of the technical support was provided from collaboration with Sweden, though this in itself was not an ideal solution. It also transpired that in previous years a considerable backlog of data errors had accumulated in the absence of checking facilities, particularly in respect of the date variables. For example, it emerged that a proportion of dates of birth had been entered unconverted, resulting in some

individuals being recorded as some seven years too old, and giving a very distorted age/sex distribution as shown in Figure 14.

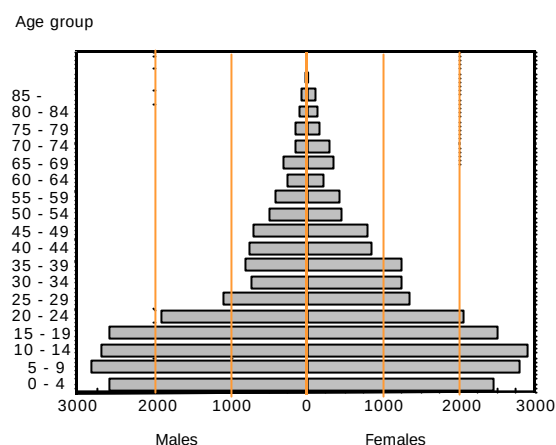


Figure 14: Distorted age-sex pyramid as a consequence of some wrongly coded dates of birth

Simultaneously implementing the new software, undertaking the re-census and trying to trace accumulated errors led to a difficult period in the life of the project which took several years to work through.

The intention from the census of 1995 was to localise data processing at the office in Butajira, so that forms would not need to be transported to Addis Ababa and real-time error detection and enquiry processes could be implemented much closer to the field. However, serious problems with the local power supply and telecommunications services made it necessary to revert to handling data in Addis Ababa, although the goal of localising it in Butajira remains, and

looks more feasible as the local infrastructure improves.

Following the extensive data cleaning processes between 1995 and 1998, it proved possible, thanks to further technological advances, to commit the definitive version of the database for the first 10 years' surveillance (1987 to 1996) to a CD-ROM. This relatively simple step has proved to be very helpful to the international group of collaborators working on the analysis of this large set of data, since all were assured of having exactly the same unmodifiable dataset to work with.

7. OUTLINING THE ANALYSIS AND DISCUSSING RESEARCH AND INTERVENTION PRIOR- ITIES

7.1 The continued surveillance

Monthly surveillance has been carried out since January 1987. The collection of data has been performed by enumerators, who were initially trained for the task. They have continuously been exposed to re-training in specific and/or new areas of work.

Vital events have been made easy for the enumerator to record. Each event is recorded on a uniquely coloured sheet. All forms are collected daily by supervisors and checked for inconsistencies and errors before they are sent to the field office.

Further attempts are being made to decentralise the processing of interview forms to the field office in Butajira town. The potential benefits of doing this are several, including closer co-operation between different types of staff. Thus, the interviewers will find it easier to discuss items with the data entry clerks. The data processing will be easier to supervise and validity and quality control processes should be more effective closer to the field. Some of the drawbacks of this decentralisation may be that electric power can be unstable and sometimes interrupted, causing a lot of computer down-time. Also, qualified researchers are not always present to supervise.

During 1999 quarterly surveillance was introduced on a pilot basis in selected areas. The purpose is to reduce the interview load on the population and to keep costs low, with minimal consequences for data quality. This pilot scheme will be carefully evaluated in the light of the third re-census, scheduled for late 1999, to compare the relative rates of missed events and other surveillance errors.

7.2 Reproductive health

In establishing the study base, the potential for reproductive health research was one envisaged application, for example in pregnancy surveillance and studies of female health and reproductive outcome.

Building on the basic population data, specific studies are being undertaken to evaluate and assess risk factors and other parameters which are associated with, or cause, poor reproductive outcomes. A prerequisite for well-designed interventions is the facility to assess morbidity, perceived health needs and social support. Focus group discussions, in-depth interviews and key informant studies are important tools. Results of quantitative multivariate risk assessments will be interpreted in conjunction with qualitative findings. Following the Safe Motherhood Initiative, providing efficient maternal health care in order to reduce maternal morbidity and mortality is a high priority. The Population Conference in Cairo in 1994 underlined women's decision making ability, property rights and literacy as important factors, thus widening the gender aspects of health. Achieving acceptable patterns of reproduction means promoting people's own choices and decisions regarding fertility, sexuality and control over reproduction in a culturally acceptable way, combined with effective strategies for reducing mortality and morbidity for women and children.

The planned project will seek to increase understanding of these inter-

related social, cultural and health issues within sub-Saharan Africa, as well as implementing and evaluating operational strategies for improvements in health care. The aim of the project is to develop and demonstrate approaches for sustainable, long-term maternal and child health programmes, promoting safe, effective and acceptable methods of reproduction. It is envisaged that the experience gained from the project would be applicable elsewhere in Ethiopia and provide a basis for an integrated approach that could be implemented more widely in sub-Saharan Africa.

The purpose of a proposed intervention study in reproductive health is to evaluate the implementation and effectiveness of specific strategies for development and aid within the reproductive health field. The aim of the interventions is to prevent bad outcomes of pregnancy. Pre- and post-intervention studies on outcomes, facilitated by the step-wedge approach, will be supplemented with cost-benefit analyses of the intervention packages. The intervention will be evaluated by different approaches, outcome research and process-based research, and using both quantitative and qualitative methodologies. The outcome research will assess the efficiency of referral by linkage of the different data records: radio calls, ambulance journeys and hospital notes. The utilisation of the referral system will be evaluated by comparing numbers of mothers from

the study area with the remaining population in the district. This process-based research is concerned to evaluate strengths and weaknesses of the chain of obstetric services, how parturients are brought into the service and how they progress through it and the interaction between staff and the parturients and their relatives as well as traditional birth attendants. Evaluation of the interventions will include the acceptance, quality and efficiency of health care. The interviews with professionals and parturients will be done using qualitative methods for evaluation. Further, social support, continuity of health care support, and women's opinions on health care will be considered. Sub-optimal care will be detected by continuous audit (including cases and controls) and by in-depth interviews following maternal and perinatal deaths as well as "near miss" episodes of, for example, life-threatening obstructed labour and postpartum haemorrhage. The continuous registration of surveillance data before and after the intervention (stillbirths, neonatal, infant, and maternal deaths), as well as using a step-wedge study design will enable evaluation of the programme's impact in the study area. It is expected that perinatal deaths could be the best proxy to evaluate the intervention. The intervention is planned to go on for a 3-year period, and the problems of sustainability and applicability for other African settings will also be considered.

7.3 ARI and indoor-air pollution

Beginning in 1999, a new applied study on Acute Lower Respiratory Infections (ALRI) and indoor air pollution is under way. Indoor air pollution has been suspected as one of several factors that affect the incidence of or mortality from respiratory infections, since in many developing countries open, unvented fireplaces are commonly used for both cooking and heating. In the few studies that report actual measurements of pollutants (mainly nitrogen dioxide and respirable particulate matter) concentrations have far exceeded threshold limit values for the working environment in most industrial countries, which in turn are considerably higher than the recommended concentration limits for the general environment [22, 23]. The epidemiologic data on indoor air pollution as a risk factor for ARI morbidity and mortality among children are ambiguous. From developing countries, several studies using proxy measures of air pollution, report an effect on ARI morbidity or mortality [6, 24-28], whereas a study, where pollution measurements are used, shows no such relation [22].

The specific background of the planned study is the finding by Shamebo, Muhe *et al* of ARI as one of the major causes of death among children in Butajira District [6] and the findings that among the factors that are most strongly related to total under-five mortality are lack of windows in the home (odds ratio 1.6), living in a traditional hut (*tukul*) and

having cooking fires in the house [7]. These factors would all be directly related to exposure to indoor air pollution, but also indirectly to other risk factors. Among children dying from ARI, the odds ratio for "having no window" is 1.8, in a logistic regression model also including factors such as ethnicity and literacy. The corresponding odds ratio for infants only is approximately 5.

The study is planned as a specialised project within the Butajira Rural Health Project. The new study is performed in collaboration with the researchers at the Community Health Department in Addis Ababa University and the Chemistry Department of the Science Faculty of the Addis Ababa University.

In 1995 a pilot study of indoor air pollution exposure was performed. The aim of the study was to make a small series of pollutant measurements in order to find the range of indoor concentrations and to study the exposure-time patterns of the children. Nitrogen dioxide, NO_2 , was measured as an indicator of indoor air pollution that can be easily quantified under field conditions. Samples are collected passively on a diffusion sampler, which precludes the need for an electric pump. Ten households in two different villages were visited for sample collection. Samples were collected in three overlapping time series, 16, 24 and 40 hours respectively. The results showed little difference between the two villages in average NO_2 concentration, but substantial between-household vari-

ation. Average concentrations ranged between $23 \mu\text{g m}^{-3}$ and $258 \mu\text{g m}^{-3}$. Conclusions from the pilot study measurements are that the between-household variation is sufficient to show differences in the children's exposure and that many of the children in Butajira are exposed to levels of NO_2 well above current European 24-hour limit values for urban air.

The exposure-time patterns of the children were studied in March 1996, by observations made by high-school students in their own homes. 120 students with siblings under five years of age, from all communities included in the study base of the Butajira Rural Health Project (BRHP) were recruited and equipped with digital watches. For three days running, they recorded the hours when a fire was burning on the hearth, and the times when siblings were indoors (hour and minute of entering and leaving the house). The number of hours the fire was burning and the total time per day of the children's smoke exposure was calculated. The number of hours per day of keeping the fire burning varied between less than 1 and more than 23, with a median value of 9 hours. The exposure times of the children varied between less than one hour (5%) and more than 15 hours (5%). The median exposure time was 5 hours. As with the NO_2 measurements, the results indicate a sufficient variation in exposure, to allow a meaningful epidemiologic study.

The study will use a case-referent design, where cases of ALRI among

under-fives will be compared to the non-case referent sample with respect to indoor smoke exposure and possible confounding factors. Exposure will be surveyed prospectively (sampling and chemical analysis plus interviews) in all households with children under five at three month intervals, as well as retrospectively (interviews) at the time of diagnosis for the case and its referents. Cases will be identified by specially trained village based health workers, who will also provide treatment for the children.

Exposure assessment will be both prospective and retrospective. In the prospective part, air pollution will be chemically measured and information on fire, cooking and children's movements will be collected in interviews. The aim of the prospective exposure survey, which will cover all households with children under five, is to characterise the household with respect to the cooking habits and to allow classification of households into groups with respect to average pollution concentrations. This will be done at three-month intervals throughout the study. Retrospective exposure assessment at diagnosis is restricted to

the households of the cases and the children in the referent sample. On identifying a case, the project health worker notifies the village based project assistant who selects referents from the BRHP population register and performs the interviews about cooking (cooking hours, what was cooked, choice of fuels) and the child's movements during cooking (indoor/outdoor). The questionnaire is being developed in collaboration with the Ethiopian researchers and will be tested in a field trial before the actual data collection starts.

In view of the enormous impact of ALRI on child health and mortality in developing countries, a contribution to primary prevention could have great value. In the more immediate future, results indicating an effect of air pollution exposure on ALRI incidence or mortality should be utilised in an intervention study in Ethiopia. Since the study is dependent on information from the households on exposure and the children's illnesses, it is planned to give a service back to the surveyed communities in the form of basic health care and a simple drug supply. The case identification method of having village based trained health workers to provide treatment for the children's respiratory infections will be extended to supply certain other drugs such as chloroquine and paracetamol to all ages.

7.4 Mental health and domestic violence

Effects of domestic violence on women's mental health have been well recognised. Victims of intimate partner violence often exhibit a variety of symptoms, including tendencies for suicide, cognitive disturbances, high avoidance or depression behaviours and high arousal or anxiety disturbances which has been conceptualised as *battered woman syndrome*, and which meets the diagnostic criteria for post-traumatic stress disorder (PTSD). Domestic violence should be viewed as a long-term, chronic condition with long-lasting emotional and psychological consequences as depression and insomnia. Gender differences of psychiatric disorders have consistently been found. A primary example is depression, which is a major threat to health and one of the most common psychiatric disorders at the community level. Gender differences have been well documented, with point prevalence, 12-month prevalence and lifetime prevalence of major depression all found in greater frequency in women than in men.

To date, population based research on mental health has not explored the extent to which domestic violence may contribute to gender differentials in depression. Epidemiological research with more accurate diagnostic techniques is needed in order to establish, at the community level, the effects of domestic violence on women's mental

health. The general objective should be to study the magnitude and characteristics of both domestic violence and depression among women of childbearing age within the Butajira research area.

Specific objectives will include examining the cultural context in which violence against women occurs, by identifying the most common beliefs and attitudes of both men and women regarding domestic violence; estimating the prevalence of domestic violence and depression/dysthymia among women of child bearing age; analysing the association between domestic violence and depression/dysthymia; determining how much violence contributes to mental health problems; and assessing the reproductive outcomes of women exposed to domestic violence.

The design of the study addressing the objectives will be population based, starting with formative qualitative research, followed by a cross-sectional survey among women, a nested case-referent study, and finally linkage of the offspring to the demographic surveillance.

7.5 Other plans

The BRHP is a continuous source of information for planning and evaluating public health interventions and a very suitable platform for research training in a developing country. The collaborating institutions, inspired by the achievements so far, are committed to maintaining the study base in the

future. Further objectives for the BRHP will be to strengthen evidence-based approaches to major public health problems in the country. It will promote and facilitate the use of research findings for planning public health interventions through timely and regular publications.

Opportunities for extending research training to the doctoral level in the Department of Community Health in Addis Ababa will be explored. The current sandwich programme for doctoral studies in Sweden constitutes a good precedent for any such undertaking in the Department of Community Health. The School of Public Health envisaged by Addis Ababa University could greatly benefit from the existence of such a well maintained "population laboratory" as the BRHP.

8. CONCLUDING REMARKS

The quality of research in Butajira can only be maintained through a collaborative undertaking that involves experienced and qualified researchers from both the quantitative and qualitative arenas in sufficient number. The collaboration

that has survived for the last 15 years between the Department of Community Health in Addis Ababa and the Department of Epidemiology and Public Health in Umeå, Sweden, is highly valued. However, there is a need to expand this collaboration in order to examine the widely diverse health problems in the country, and to expand training opportunities in various specialised field such as nutrition and health education. One of the major priority areas for new research is to study the epidemiology of HIV/AIDS in rural setting and the spread of infection from urban to rural settings.

The major challenge for sustaining this study base would be to get continuous funding for its activities. Plans for sustainability include developing a cost-sharing scheme (in which projects served by the study base contribute a certain proportion of their budget), diversifying funding from external sources, and soliciting local funding.

In conclusion, the BRHP is a national asset that can provide scientifically sound information for use in public health planning, policy making and evaluation. It also provides a fertile platform for training in epidemiology and public health for the country.

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